



SWE 4603

Software Testing and Quality Assurance

Week 8

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Lesson Outcome

- Automation Testing

Week on:

- Chapter 15: Automation and Testing Tools

Automation Testing

Automation Testing

What is Automated Software Testing?

- ❖ is the methodology that helps to validate the functioning of the software before it is moved to production.
- ❖ In this process, **automated testing tools** are used by the QA teams for executing the test scripts.
- ❖ With the use of automated software testing tools, QA teams can quickly test the software, prepare the defect reports, and compare the software results with the expected results.

Automation Testing

What is Automated Software Testing?

- ❖ automated testing process provides several benefits
 - faster delivery
 - eases regression testing time
 - ensures quality software along with reducing manual testing efforts.
- ❖ Automated testing ensures faster and quality releases utilizing automated testing tools.
- ❖ It is actually a process that validates if the software is functioning appropriately and meeting the given requirements using the test automation tools.

Automation Testing

Why is automated software testing important?

- ❖ Automated software testing uses **open source/paid test automation tools** to test applications quickly and more efficiently 24×7 without any human intervention.
- ❖ It takes up different testing approaches and replaces several tasks of manual testers and ensures to improve the overall test coverage effectively.
- ❖ Pre-written scripts are tested to ensure quality software is released and essentially reduces the regression testing time.
- ❖ Test automation also helps enterprises to improve the communication between DevOps, QA teams, and product owners who depend on the results of these automated tests, as these results are more reliable than the manual testing results.
- ❖ It helps QA teams to re-check and validate the outcomes quickly and more effectively with less human effort.

Automation Testing

What are the benefits of Automated Software Testing?

The benefits have been broadly classified as

1. **Application-Wise benefits**
2. **Cost-wise benefits.**

Automation Testing

Application-Wise benefits

- ❖ **Increases test coverage:**
 - helps to perform more number of automated test scripts and increases the test coverage and the scope of tests to enhance the applications quality.
 - It also helps in saving time and significantly reduces manual testing efforts.
- ❖ **Ensures test accuracy:**
 - Manual test cycles might lead to errors, but with automated software testing tools, there is an assurance that the testing practice and validation of the application are performed with good accuracy, as errors are identified at every phase.
- ❖ **Eases regression testing time:**
 - lot of time, and human efforts can be saved when automated regression testing is practiced with testing tools and manual testers get time to be used for other quality works.
- ❖ **Facilitates with reusable test scripts:**
 - it eases the tester's efforts as the test scripts can be reused with some minor changes made in the test scripts.
 - Also, these scripts can be used multiple times as they can be stored, and can be re-used for repeating the tests.

Automation Testing

Cost-Wise Benefits

❖ **Saves Cost:**

- With automated app testing, enterprises can effectively save their budget as the test cases are run at a faster speed, defects are identified early and fixed before it is moved to production.
- Though the initial costs of automation are high, but when once the automated test framework is developed, it eases the testing tasks and remarkably saves the overall costs.

❖ **Increases ROI:**

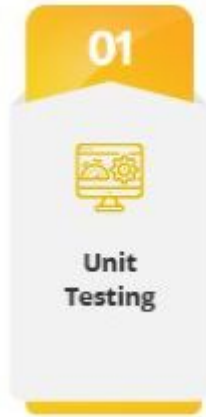
- With the development of an effective test automation framework, returns are huge as testing is performed faster with the inbuilt features.
- Moreover, automation testing delivers faster and quality results, improves the time to market and finally ensures increased return on investment.

❖ **Saves Time:**

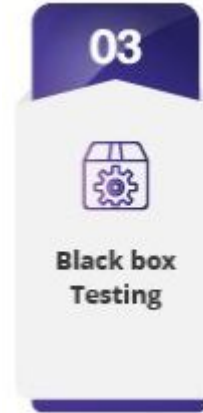
- As tests are run automatically 24×7, automated software testing helps to save time.
- Moreover, with automated testing tools, manual scripts are automated and the regression testing time will be drastically reduced.
- This eventually leads to saving the delivery time of the application and largely saves lot of manual efforts and saves time.

Automation Testing

Automation testing in use



These tests are usually written by the developers, but sometimes automation testers also write them. These tests are usually performed during the software development phase.



is performed to check if there are any incorrect or missing functions, checks for the existence of any interface errors or any behavior-related or performance errors, and also checks the presence of termination errors, if any.



to check if all the functions work as expected or not. This involves testing APIs, databases, and security and also tests the overall functionality of the application.
Regression testing:

Automation Testing

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**Smoke
Testing**

This type of testing is performed to check whether the build is stable or not. If the build is not stable, then it will be sent back to the developers as unstable build, and if the build is stable, then testers will accept it to perform further testing on the build

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**Regression
Testing**

to check whether all the existing features are functioning well even with recent code changes. It basically verifies that the old code works in the same way as it did before making any changes to the code

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**Integration
Testing**

In this integration testing, the application modules are typically integrated logically and tested as a group. checks and verifies the data communication between different modules of the application.

Automation Testing

What Does an Automation Tester Do?

- ❖ The QA automation tester is responsible for designing and writing test automation scripts as the first phase and they use various types of test automation frameworks to achieve successful test automation.
- ❖ They run test scripts to ensure all the scenarios are successfully tested and they have a close association with the developers and other QA analysts.
- ❖ They log bugs and issues in various test management tools such that the developers work on these bugs and fix them.
- ❖ Once again automated regression testing is done to ensure the new functionalities do not affect the functionalities of existing features.

Automation Testing

What are automation testing challenges, and how do you overcome them?



Automation Testing

Following are some facts pertaining to the cost incurred in testing tools:

- ❖ **Automated script development:** Automated test tools do not create test scripts. Therefore, a significant time is needed to program the tests.
- ❖ **Training is required:** It is not necessary that the tester be aware of all the tools and be able to use them directly. He may require training regarding the tool; otherwise, it ends up on the shelf or implemented inefficiently
- ❖ **Configuration management:** Configuration management is necessary to track a large number of files and test-related artifacts.
- ❖ **Learning curve for the tools:** There is a learning curve in using any new tool. For example, test scripts generated by the tool during recording must be modified manually, requiring tool-scripting knowledge in order to make the script robust, reusable, and maintainable.
- ❖ **Multiple tools are required:** It may be possible that your requirement is not satisfied with just one tool for automation. In such a case, you have to go for many tools, which incur a lot of cost.

Automation Testing

Tools for automation testing

- ❖ Katalon Studio
- ❖ Applum
- ❖ Selenium
- ❖ Cypress

Automation Testing

Selection Criteria for tools

- ❖ **Match the tool to its appropriate use:** Before selecting the tool, it is necessary to know its use. A tool may not be a general one or may not cover many features. Rather, most of the tools are meant for specific tasks.
- ❖ **Select the tool for its appropriate SDLC phase:** Since the methods of testing change according to the SDLC phase, the testing tools also change.
- ❖ **Select the tool according to the skill of the tester:** The individual performing the test must select a tool that conforms to his skill level. For example, it would be inappropriate for a user to select a tool that requires programming skills when the user does not possess those skills.

Automation Testing

Selection Criteria for tools

- ❖ **Select a tool that is affordable:** Tools are always costly and increase the cost of the project. Therefore, choose the tool that is within the budget of the project.
- ❖ **Determine how many tools are required for testing the system:** A single tool generally cannot satisfy all test requirements. It may be possible that many test tools are required for the entire project.
- ❖ **Select the tool after examining the schedule of testing:** First, get an idea of the entire schedule of testing activities, and then decide whether there is enough time for learning the testing tool and then performing automation with that tool. If there is not enough time to provide training on the tool, then there is no use of automation.